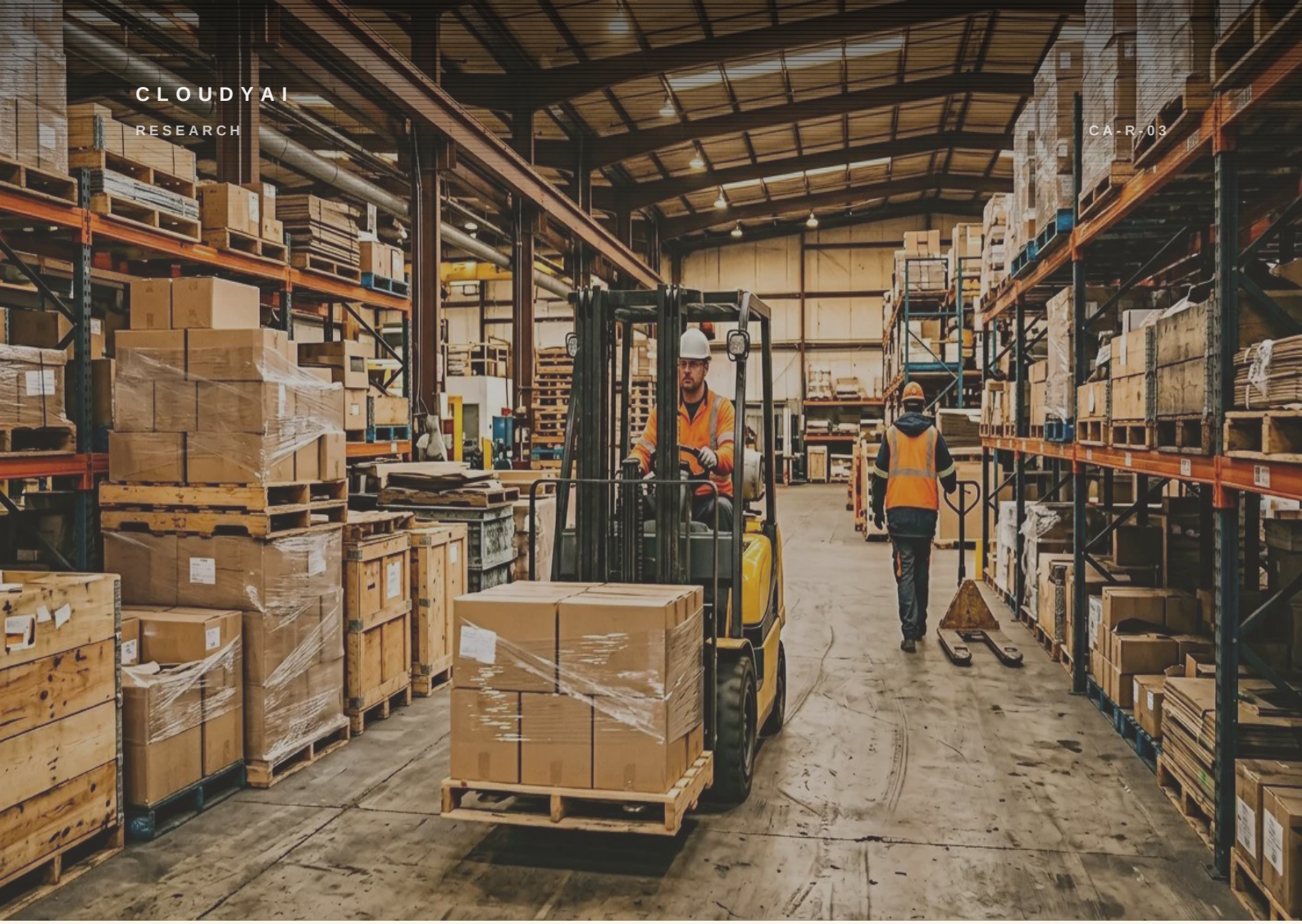


COMPUTER VISION

Counting What You Can See

Camera-derived OEE for production lines that never got sensors.

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SUMMARY

Most lines have a photo-eye at the end and a clipboard at the start. A camera can recover the numbers in between: micro-stops, changeover time, queue length, and the difference between blocked and starved.

Ask a shift supervisor how the line ran yesterday and you will get a story. Ask for the numbers and you will get a production report that counts what came out of the door, not what happened between seven and three. The gap between those two things - the stops, the slow cycles, the changeover that took forty minutes instead of twenty - is where most of the recoverable capacity sits.

OEE, availability times performance times quality, is a good frame. The problem is instrumentation: most small and mid-sized plants have a photo-eye at the end of the line, a tally on a whiteboard, and no idea what happened in the middle.

Why a camera is a reasonable sensor here

- It is already installed, or it costs less than a new sensor network to install.
- It captures context - a jam, a missing operator, a pallet blocking the exit - not just a count.
- It does not care where you put the counter later; you can move the counting line in software without touching a cable.
- It can distinguish product variants, which a beam break cannot.
- It produces a clip, which ends arguments about whether the line stopped at 10:40 or 11:10.

The four numbers worth having first

Units in and out

Detect the unit, track it across frames, count it when it crosses a line. Accuracy of 97 to 99 per cent is achievable under stable lighting with a fixed camera and a consistent product. That sounds impressive until you multiply by a million units a month - so always reconcile against the production report daily and treat a two per cent drift as a signal that something in the scene changed.

Cycle time distribution, not average

An average cycle time hides the two long stops that ruined the shift. A camera gives you a timestamp per unit, which gives you a distribution: the normal cycle, the slow tail, and the gaps. The tail is where the money is.

Micro-stops

Stops under five minutes rarely get written down. They are also, in aggregate, frequently the largest single availability loss on a line. A camera that logs every gap over 20 seconds turns an invisible loss into a ranked list.

Changeover duration

The green bottle ends at 14:02 and the amber bottle starts at 14:41. That is a 39-minute changeover, evidenced, comparable across shifts and crews, and impossible to argue with. Once you can see it, you can start reducing it.

Turning detections into line states



A dispatch aisle. Counting, queue length and blocked states are all visible from one fixed view.

The useful abstraction is a small state machine per line. Each frame is classified into one of a handful of states, with temporal smoothing so a single dropped detection does not flap the state.

State	Visual evidence	What it usually means
Running	Units moving across the counting line at normal cadence	Fine
Starved	Counting line empty, upstream section empty	Upstream machine or material feed
Blocked	Units queued and stationary at the discharge	Downstream bottleneck or pallet exchange
Stopped	Line full of stationary units, operator present	Breakdown, adjustment, cleaning
No product	Line empty, operator absent	Changeover, pre-shift, planned stop

Attribution rules should be written down and reviewed with the production team, because the same visual state can have two legitimate causes. The camera proposes; the supervisor confirms. Systems that assign blame automatically get unplugged.

A dashboard a supervisor will actually use

Three numbers on one screen, updated live: current rate against target, minutes lost so far this shift, and the top three stop reasons with clips attached. Everything else on a second screen, available but not competing for attention. If the screen needs training to read, it will not be read at 06:20 on a Monday.

> **Design constraint**
> The dashboard must load on a phone on the plant Wi-Fi. If it needs a laptop and a VPN, it will be opened once a month by somebody in the office, and the shift team will keep using the whiteboard.

Accuracy, drift, and the reconciliation habit

Any camera-based count will drift as the scene changes: a new product colour, a moved pallet, a lens that fills with dust over a Harmattan or a dusty dry season. Build three cheap habits. A daily comparison against the production report with a tolerance band. A weekly review of the highest-confidence and lowest-confidence detections. A monthly re-check of ten minutes of footage by a human. When the count drifts, re-label a few hundred frames from the current scene and fine-tune; this is routine maintenance, not a project.

A six-week pilot

01 Week 1: choose one line and one counting point. Pull a week of footage and measure accuracy offline before touching production.

02 Week 2: add the state machine and the stop-reason taxonomy with the supervisor.

03 Weeks 3-4: shadow mode. Run the system next to the existing report; publish both numbers; investigate every disagreement.

04 Week 5: switch primary reporting to the camera, keep the manual count as a cross-check.

05 Week 6: review with the team, agree thresholds, hand over the runbook, decide whether to expand.

What the numbers are worth

Illustrative only: a line running 4,000 units per shift, 250 shifts a year, with a 4 per cent margin per unit, recovering 6 per cent of lost availability through better visibility is worth more than the system costs in the first year on most lines we look at. The conservative version of that statement is simpler: if the camera finds one recurring stop reason per month that nobody had named, it has paid for itself. Do the arithmetic with your own numbers before believing anyone's case study, including this one.

Limits

A camera cannot measure torque, temperature, or pressure. It sees what is visible from where it hangs, and it will miss anything that happens behind a guard or inside a closed machine. It is a complement to PLC data, not a replacement, and on a fully instrumented line the marginal value is much lower. Spend the money on the line that has no data at all.

ABOUT THIS PAPER

CloudyAI is a Zimbabwean AI studio. We deploy computer vision on existing plant cameras, audit AI spend, and help teams respond when an AI system misbehaves. Research is published free, without an email wall.

Corrections and questions: research@cloudyai.co.zw. If you would like this method applied to your plant, the first conversation is free and the camera audit is a fixed-price week.